GATE 2027 / QUESTION PAPER AUDIT

2017-2026 coverage and gaps

A syllabus-first audit for the ten Computer Science sections

What this report does

Compares the supplied IIT Madras GATE 2027 CS syllabus and the current beginner-note lesson map with official GATE CS paper archives. It identifies concepts already represented, deeper skills the notes needed, and adjacent topics that should not displace the 2027 syllabus.

The requested scope

Ten technical sections: Engineering Mathematics, Digital Logic, COA, Programming and Data Structures, Algorithms, Theory of Computation, Compiler Design, Operating Systems, Databases, and Computer Networks. General Aptitude is a separate common section and is kept outside this ten-subject audit.

How to read the question anchors

Examples are tagged by paper session (CS1/CS2 where applicable) and question number. A question can test multiple skills. A match means the notes teach the underlying method, not that they repeat the exact question. The 2025 session label matters: question numbers differ between CS1 and CS2.

Trust and scope note

The pasted Gemini conversation was treated as context, not as evidence. Its claimed missing-topic lists and question mappings were checked against actual paper text where available. Older archive files include image-only scans and one malformed 2017 Set 1 copy; for those, this audit reports recurring topic coverage instead of claiming a machine-verified question-by-question transcript.

01 / SUMMARY

What changed in the notes

All 2027 syllabus headings remain the study spine; PYQs sharpen how each concept is used.

MATH	**Matrix powers / rank-nullity / integration by parts**	Added worked method lessons
DL	**K-map, mux and state tracing / signed encodings**	Added step-by-step circuit and representation lessons
COA	**IEEE fields / nested AMAT / hazard timing**	Added formulas with assumptions and worked arithmetic
PDS	**C pointers / heap indices / tree invariants**	Added memory tracing and shape rules
AL	**Recurrences / BFS-DFS / MST / hashing**	Added selection rules and edge-case checks
TOC	**Closure / modulo DFA / ambiguity / decidability**	Added constructions, state equivalence and boundaries
CD	**GEN-KILL / fixed points / backpatching**	Added exact equations and evaluation sequence
OS	**Banker safety / paging / TLB**	Added algorithm trace and address split
DB	**FD closure / schedule recovery / B+ trees**	Added separate correctness tests
CN	**Window bounds / fragmentation / longest prefix**	Added protocol and numeric workflows

Important: these are teaching supplements, not a promise that any future paper will repeat a past pattern. Learn the syllabus concept so new numbers and stories remain solvable.

01 / MATH

Engineering Mathematics

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors across years: 2018 Q17/Q26 (eigenvalues/matrices), Q28 (logic), Q44 (probability); 2022 Q34/Q36/Q45 (limits, generating functions, LU); 2023 Q17/Q27/Q30 (determinants, calculus, integration); 2024 CS1 Q11/Q12/Q49/Q63; 2025 CS2 Q11/Q12/Q14/Q42/Q44; 2026 CS1 Q13/Q20/Q32/Q58.

Gap closed in this revision

Added full worked lessons for matrix powers/Cayley-Hamilton, rank-nullity, integration by parts, conditional probability and independence.

Syllabus boundary

Not a separate 2027 topic: advanced numerical-analysis methods beyond the listed calculus outcomes. Prioritize the listed limits, derivatives, extrema, MVT and integration.

Where the new teaching is placed

New lesson pages: Matrix powers, eigenvalue facts and nullity; Integration by parts and probability traps. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Digital Logic

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q22/Q33/Q49 (sequential logic, fixed point, Boolean minimization); 2022 Q18/Q40-Q41 (signed arithmetic, display/control and IEEE representation); 2023 Q20/Q42-Q44 (muxes, sequential circuits, Boolean logic and floating point); 2024 CS1 Q13/Q28/Q47/Q64; 2025 CS1 Q24/Q42/Q60 and CS2 Q31/Q34/Q43/Q50; 2026 CS1 Q21/Q22/Q37/Q48.

Gap closed in this revision

Added explicit K-map grouping, mux-as-function, counter state tracing, sign extension, signed overflow and IEEE field decoding.

Syllabus boundary

The 2027 syllabus includes fixed and floating-point number representation; deeper circuit fabrication details are not listed.

Where the new teaching is placed

New lesson pages: K-maps, multiplexers and state tracing; Signed arithmetic and floating-point decoding. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Computer Organization and Architecture

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q23/Q34/Q50-Q51 (DRAM, cache mapping, pipeline and instruction encoding); 2022 Q17/Q33/Q54 (DMA and cache performance); 2023 Q32/Q41/Q63 (pipeline, memory interfacing, cache); 2024 CS1 Q15/Q30/Q53-Q54; 2025 CS2 Q28/Q32/Q39/Q49/Q55-Q56; 2026 CS1 Q14-Q16/Q36/Q38/Q60.

Gap closed in this revision

Added cache address derivation, nested AMAT, write policies, IEEE special values, RAW hazards, forwarding and pipeline clock-time calculation.

Syllabus boundary

Cache coherence and detailed multicore protocols are not named in the supplied 2027 COA syllabus.

Where the new teaching is placed

New lesson pages: Cache address fields and multi-level AMAT; IEEE 754 numbers, hazards and pipeline time. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Programming and Data Structures

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q20-Q21/Q29/Q46 (trees, C recursion/pointers and heaps); 2022 Q15/Q21/Q28/Q43/Q62 (lists, C, BSTs and stack/queue); 2023 Q11-Q12/Q34/Q45-Q46/Q58; 2024 CS1 Q18-Q19/Q41/Q43; 2025 CS1 Q26/Q34-Q35/Q62 and CS2 Q13/Q19/Q33/Q35/Q45; 2026 CS1 Q23/Q33-Q34/Q39-Q40/Q62 and CS2 Q12/Q49-Q50/Q60-Q61.

Gap closed in this revision

Added C pointer arithmetic, array/pointer equivalence, static lifetime, undefined sequencing warning, full binary tree identities, heap index formulas and BST inorder invariant.

Syllabus boundary

The 2027 list specifies C and core structures; language features outside C and implementation-specific compiler behavior are not core.

Where the new teaching is placed

New lesson pages: C pointers, arrays and variable lifetime; Tree shape, heaps and traversal invariants. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Algorithms

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q31/Q45/Q47-Q48 (dynamic programming, recursion, MST and knapsack); 2022 Q11/Q16/Q49/Q51 (complexity, hashing, MST and recurrence); 2023 Q14/Q19/Q28/Q45/Q53 (recurrence, hashing, asymptotic growth and heaps); 2024 CS1 Q17/Q34/Q42/Q45; 2025 CS1 Q18/Q20/Q33/Q43/Q64-Q65 and CS2 Q20/Q29/Q37/Q41/Q59; 2026 CS1 Q17/Q24/Q41/Q49-Q50 and CS2 Q24-Q25/Q32/Q36-Q39.

Gap closed in this revision

Added recurrence solving, greedy exchange intuition, BFS layers, DFS components, MST cut property, shortest-path assumptions and hashing traces.

Syllabus boundary

Amortized analysis, SCC, articulation/bridges and max-flow are not explicitly listed in the supplied 2027 Algorithms syllabus; treat as optional extensions unless an official syllabus update adds them.

Where the new teaching is placed

New lesson pages: Solving recurrences and proving a greedy choice; Graph search, shortest paths and minimum spanning trees. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Theory of Computation

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q35-Q37/Q52 (language classes, decidability, lexical regular expressions); 2022 Q12/Q23/Q46-Q48 (regular expressions, closure and decidability/CFLs); 2023 Q13/Q23/Q38-Q39/Q62; 2024 CS1 Q23/Q50/Q61; 2025 CS2 Q24-Q25/Q30/Q52/Q60; 2026 CS1 Q25-Q26/Q51-Q52 and CS2 Q13/Q29/Q47-Q48.

Gap closed in this revision

Added product automata, modulo counters, state distinguishability, CFG ambiguity, PDA power, decidable-vs-undecidable checks and pumping-lemma limits.

Syllabus boundary

Do not use the pasted Gemini mapping without the session code: its Q30 closure claim refers to 2025 CS2; 2025 CS1 Q30 is a counting problem about consecutive equal symbols.

Where the new teaching is placed

New lesson pages: Regular languages: closures, products and minimal DFAs; CFG questions: parse trees, PDA power and decidability. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Compiler Design

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q37-Q38 (lexical analysis and parsing); 2022 Q13/Q29/Q65 (LALR/LR and syntax-directed translation); 2023 Q18/Q59 (lexical tokens and translation); 2024 CS1 Q26/Q37/Q39; 2025 CS1 Q12-Q13/Q46/Q52 and CS2 Q21-Q22/Q40-Q41/Q51; 2026 CS1 Q27-Q28/Q42/Q53 and CS2 Q35/Q41/Q45.

Gap closed in this revision

Added precise GEN/KILL equations for reaching definitions, available expressions and liveness, fixed-point iteration, dependency ordering, leaders and backpatching.

Syllabus boundary

The user-supplied Gemini answer described several 2025 CD question types incorrectly; the listed verified anchors match the actual session PDFs.

Where the new teaching is placed

New lesson pages: Data-flow analysis: GEN/KILL and fixed points; Attribute dependencies, backpatching and basic blocks. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Operating Systems

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q24/Q39-Q40 (resource allocation and producer-consumer); 2022 Q19/Q26/Q38/Q42/Q64 (semaphores, deadlock, TLB, scheduling and LRU); 2023 Q21/Q26/Q37/Q56-Q57 (context switch, scheduling, synchronization and paging); 2024 CS1 Q24-Q25/Q40/Q62; 2025 CS1 Q29/Q38/Q54 and CS2 Q26-Q27/Q47-Q48; 2026 CS1 Q29/Q35/Q63-Q64 and CS2 Q23/Q51/Q55.

Gap closed in this revision

Added Banker safety check, Need/Available matrices, virtual-address splitting, TLB paths and explicit replacement tracing.

Syllabus boundary

System-specific kernel internals beyond calls, processes, synchronization, scheduling, memory and files are not listed.

Where the new teaching is placed

New lesson pages: Deadlock avoidance and Banker safety; Virtual addresses, TLBs and page replacement. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Databases

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q11-Q12/Q41-Q42 (ER, SQL, relational algebra and dependencies); 2022 Q14/Q25/Q31/Q39/Q56 (normal forms, SQL, keys and schedules); 2023 Q15/Q60-Q61 (relations, SQL and file indexing); 2024 CS1 Q20-Q22/Q35/Q44/Q46; 2025 CS1 Q15/Q21/Q39/Q47/Q55 and CS2 Q27/Q46/Q53-Q54/Q57; 2026 CS1 Q30-Q31/Q43/Q65 and CS2 Q15/Q20/Q42/Q46/Q53.

Gap closed in this revision

Added attribute closure and key checks, lossless-vs-dependency-preserving distinction, recoverability/cascadeless/strict schedules, and B+ splits.

Syllabus boundary

4NF/5NF and ARIES recovery are not named in the supplied 2027 database syllabus; keep as optional extension after the listed normal forms and transactions.

Where the new teaching is placed

New lesson pages: Functional dependencies, keys and decomposition checks; Schedules: serializability, recovery and B+ tree operations. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Computer Networks

Verified anchors and additions for GATE 2027 preparation

Question anchors

Verified anchors: 2018 Q13-Q14/Q25/Q54-Q55 (protocol fields, TCP congestion, sequence numbers, fragmentation and MAC); 2022 Q22/Q35/Q55/Q57/Q60 (Ethernet, DNS, routing and sequence numbers); 2023 Q16/Q24/Q49/Q51/Q64; 2024 CS1 Q16/Q29/Q31/Q36/Q58/Q65; 2025 CS1 Q16/Q22/Q40/Q57 and CS2 Q16-Q18/Q23/Q36; 2026 CS1 Q18-Q19/Q44-Q45/Q56 and CS2 Q21-Q22/Q33/Q43/Q58/Q65.

Gap closed in this revision

Added bandwidth-delay window intuition, sequence-number bounds, longest-prefix routing, NAT, IPv4 fragmentation and TCP flow-vs-congestion windows.

Syllabus boundary

The supplied 2027 syllabus explicitly lists IPv4, distance-vector/link-state routing, TCP, DNS and HTTP; IPv6 and BGP are optional extras, not core coverage.

Where the new teaching is placed

New lesson pages: Reliable delivery, windows and link efficiency; IPv4 delivery: CIDR, NAT and fragmentation. Each added page starts with intuition, a diagram, a worked example, step-by-step concepts, an exam trap and a recall prompt.

Corrections to the pasted conversation

Use actual session paper labels when studying a question number.

2025 TOC mapping

The pasted text mixed the two 2025 sessions. CS2 Q30 is the closure-properties matching question; CS1 Q30 counts ternary strings containing a repeated adjacent symbol. CS2 Q25 is CFG decidability, Q51 is LL(1)/dangling-else analysis, Q52 tests regularity of modular/counting languages, and Q60 asks for a minimum-DFA state count for a product-mod-7 language.

2025 Mathematics mapping

The matrix-power, integration-by-parts, determinant, induction/predicate, lattice and LU examples in the pasted conversation map to CS2 Q11, Q12, Q14, Q15, Q42 and Q44 respectively. The matrix-power shortcut was added because the previous lesson named matrices but did not explicitly teach repeated powers/Cayley-Hamilton.

Unverified claims kept out of the core

The Gemini list proposed SCC/max-flow, 4NF/5NF/ARIES, IPv6/BGP, cache coherence, and other advanced additions without tying each claim to an official question. These subjects are not explicitly named in the supplied 2027 syllabus. The notes label them as optional extensions instead of treating them as mandatory 2027 preparation.

Syllabus source

The supplied 2027 syllabus contains ten technical sections. The official GATE archive pages identify previous-year master papers and separate sessions for relevant years. This report records selected verifiable question anchors rather than reproducing copyrighted question text.

13 / YEAR COVERAGE

Which years are verified at question level

Question-number anchors appear only where paper text could be checked reliably.

2017-2021 / source limitation

The official archive copies for these years include scans with no reliable text extraction; the local 2017 Set 1 copy is malformed. Available 2017 Set 2, 2019, 2020 and 2021 scan files were reviewed, but OCR did not recover enough question text to certify question-by-question mappings. These years are not represented as fully audited in the subject anchors.

2018 / verified paper text

The archived 2018 CS paper was text-readable. Selected question anchors were checked and are listed in each relevant subject crosswalk.

2022-2023 / verified paper text

The archived papers were text-readable. Selected question anchors were checked against the paper statements and mapped to the relevant syllabus skills.

2024-2026 / session-aware verification

Official digital papers were checked, with CS1 and CS2 question numbers kept distinct where multiple sessions were held. The crosswalks list selected anchors, not every question.

How to interpret a missing anchor

No anchor for a year or topic does not mean “not asked” or “out of syllabus.” It means this report does not certify that question number from a reliable paper transcript. The notes still teach the supplied 2027 syllabus across all ten technical subjects.